
Neighborhoods and Segregation

LSE GY457 - AT 2025

Topic 8

Sara Bagagli

Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- Topic 4&5: Monocentric city model
- Topic 6: Firm location choice

- Block 3

- Topic 7: The vertical dimension of cities
- **Topic 8: Neighborhoods and segregation**
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Recap from last time

- Vertical rent gradient
 - There is a ground floor premium driven by retail
 - Positive price gradient at higher floors driven by offices
- Can height elasticities be used to find profit-maximizing building height?
 - Marginal average unit cost equals marginal average unit rent
 - Determine maximum bid for a given internal rate of return
- Spatial equilibrium
 - Skyscrapers are both a cause and an effect of urbanization
 - Height regulations lead to welfare loss, but not necessarily higher rents

Today: Neighborhoods and segregation

- Housing and neighborhood characteristics
 - Urban housing and hedonic models
 - Impact of neighborhood characteristics on property prices
 - Housing demand and discrimination in lending market
- Theories of segregation
 - Rise and decline of the American ghetto
 - Causes of segregation: White flight (Boustan, 2010)
 - Race-based tipping dynamics
- Impacts of transportation infrastructure on segregation
- Consequences of segregation and its implications
 - Role of ethnic enclaves

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Urban housing and neighborhood choice

- Housing fundamental part of people's lives and of urban areas
- Housing is very durable and immovable
 - High degree of durability means that market for housing dominated by existing units
 - Immovability means that the choice of a particular housing unit embeds the choice of the residential location (which includes neighbors, school quality, centrality, crime rate,..)
- Neighborhood choice
 - When household choosing a house, choosing more than dwelling
 - Choose set of local public goods and set of (local) taxes to finance

Estimating housing prices: Hedonic model

- Main idea:
 - A house consists of many characteristics, each with a price
 - The market price of a house is the sum of the market prices of each of its components
- How can we estimate the housing price based on this model?
 - Regress house price on the set of characteristics
 - Characteristics may include: size, number of bedrooms, floor level, presence of garden/balcony, year of construction,...
 - e.g.: $\ln(\text{HouseValue}) = a_0 + a_1 \text{Size} + a_2 \# \text{Beds} + a_3 \text{Garden} + a_4 \text{Tube} + \varepsilon$
 - Helpful to indicate the market value of particular attributes

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Impact of neighborhood characteristics on property prices

- Price of house reflects attributes of the house *and* the characteristics of the neighborhood in which the house is located
- One aspect has drawn a lot of attention in the literature: the racial composition of neighborhoods
 - Ambiguous effects (more on this later)
- *Which neighborhood characteristics may affect the price of a house?*

Impact of neighborhood characteristics on property prices: Literature findings

School quality	↑	Heavy traffic on street	↓	Proximity to rail lines, highway, transit lines	↓
Crime rate	↓	Location in floodplain	↓	Proximity to highway interchanges	↓
Property tax rate	↓	Dist to employment	↓	Proximity to church	↓
Neighborhood income and proxies (e.g., nhb house size)	↑	Dist to shopping	↓	Existence of zoning or restrictive covenants	↑
Air pollution	↓	Dist to airport	↓	Previous house price increase in neighborhood	↑
Airport noise	↓	Rating of house quality next door/block	↑	Proximity to contaminated areas	↓
Proximity to park	↑	Within walking distance to public transit	↑	Proximity to nuclear power plant	↓
Industrial noise	↓				

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Housing demand and discrimination in lending market

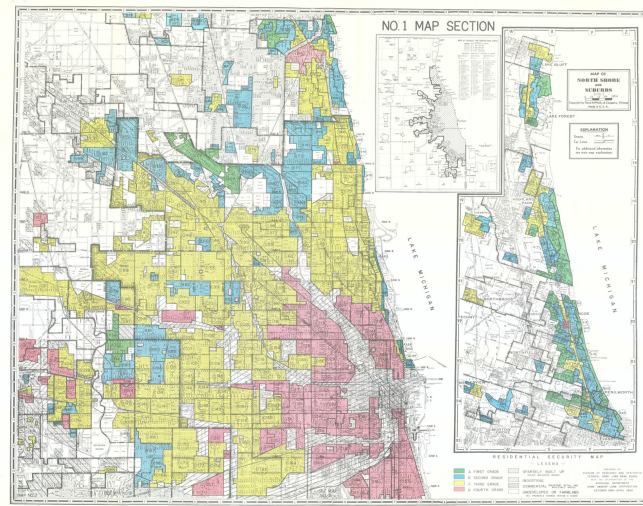
- Large ethnic differences in homeownership rates in US
 - In 2001, 74.3% of white households owned their homes vs. 47.7% of African Americans and 47.3% of households of Hispanic origin
- Reasons include aspects like income differences and differences in credit risk, but also discrimination in access to mortgage credit
 - In the 1930s, the Home Owners Loan Corporation (HOLC) produced a series of highly controversial maps aimed at documenting neighborhood-specific riskiness of mortgage lending, based on house and neighborhood characteristics (including racial and ethnic composition)

Redlining

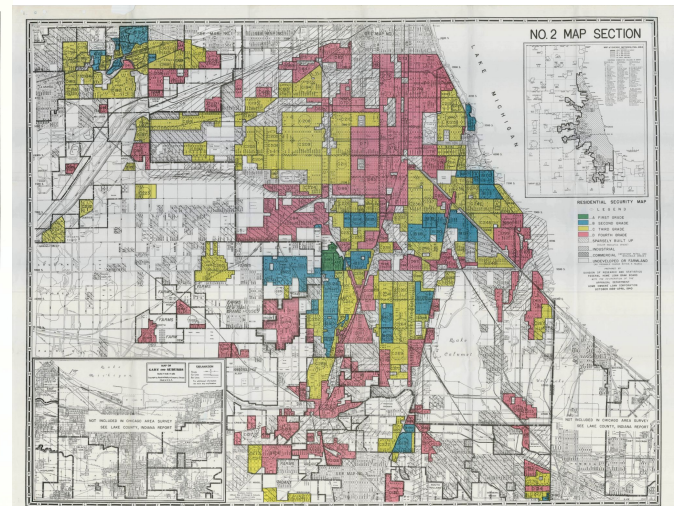
- “Practice of denying a creditworthy applicant a loan for housing in a certain neighborhood even though the applicant may otherwise be eligible for the loan”
 - Refers to the presumed practice of mortgage lenders to drawing red lines around portions of a map to indicate areas or neighborhoods in which did not want to make loans
 - HOLC maps gave grades A, B, C, D to neighborhoods, with A being the “best” and D being the “hazardous” (colored in red, hence the term *redlining*)
 - Were guidelines (not legal requirements), but found to have long lasting negative effects on house values (Aaronson et al., 2021)

Redlining: Chicago example

Chicago HOLC Map (North side)



Chicago HOLC Map (South side)



Source: University of Richmond, Mapping Inequality (<https://dsl.richmond.edu/panorama/redlining>)

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Rise and decline of the American ghetto (Cutler et al., 1999)

- Studies segregation in US cities between 1890-1990
- Question: How are neighborhoods formed?
 - Evidence showing that neighborhoods are important determinants of economic outcomes
- History of segregation can be divided into three periods
 - 1890-1940: Birth of the urban ghetto
 - 1940-1970: Expansion of the ghetto
 - 1970-1990: Decline of the ghetto
- In 1970, average African American lived in nhbd that was 68% Black; in 1990, in nhbd that was 56% Black

Rise and decline of the American ghetto (Cutler et al., 1999)

- Why this pattern of segregation developed?
- Three classes of theories of segregation
 - *Port of entry theory*: Minorities prefer to live among members of their own race
 - *Centralized/Collective action racism*: White use legal, quasi-legal, or violent and illegal barriers to keep Black out of White nhbds
 - *Decentralized racism*: White segregate themselves by paying more to live with members of their own race
- All theories predict segregation should change rapidly with incoming of new migrants into city

Rise and decline of the American ghetto (Cutler et al., 1999)

- How do these theories of segregation relate to the housing costs for Black and White households?
 - Which group do we expect to be paying more for housing under each theory and why?
- Idea: Consider how the Black-White housing price difference varies across cities as the level of segregation varies to understand the source of segregation
 - Ex ante, indeterminate how changes in discrimination affect differences in Black and White housing costs (depends on strength of homophily, and on source of segregation)

Rise and decline of the American ghetto (Cutler et al., 1999)

- Tests for the sources of segregation:
 - Segregation resulting from centralized White racism or from desire of Black to live with others of same race will increase housing costs for Black relative to White households (*why?*)
 - Segregation resulting from decentralized White racism will reduce housing costs for Black relative to White (*why?*)
- Further differentiate between centralized White racism hypothesis and port of entry hypothesis
 - How? Looking at housing prices for different groups of Black population (new migrants into city vs. long-term residents)
 - If port of entry: Black migrants pay more than Black residents

Rise and decline of the American ghetto (Cutler et al., 1999)

TABLE 6
PREDICTIONS OF ALTERNATIVE THEORIES

RELATION BETWEEN SEGREGATION AND:	THEORY		
	Port of Entry	Collective Action Racism	Decentralized Racism
House prices	Blacks pay more (esp. migrants)	Blacks pay more	Whites pay more
Attitudes toward integration	Blacks prefer seg- regation (esp. migrants)	Whites prefer seg- regation	Whites prefer seg- regation

Rise and decline of the American ghetto (Cutler et al., 1999)

- Main findings:
- In 1950s, higher levels of segregation resulted from collective actions from White to exclude Black (e.g., restrictive covenants, violence)
 - Result: Black households in more segregated cities paid more for housing than they would have
- By 1990, segregation stems from decentralized racism, with White preferring to live in White neighborhoods and paying more for housing
 - Result: Segregation premium paid by Black has reduced

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Causes of segregation: White flight

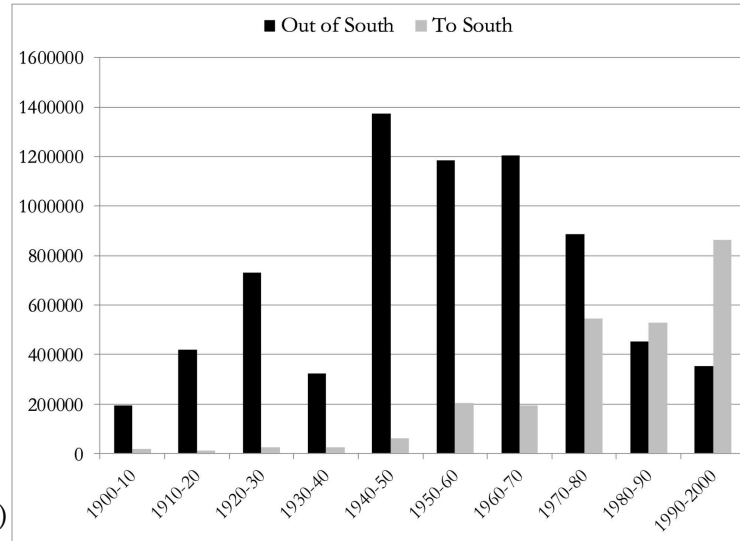
(Boustan, 2010)

- “Was postwar suburbanization white flight? Evidence from the Black Migration”
- *White flight*: process by which White households left central cities to avoid living in racially diverse nhbs
- Paper documents that Black migration reduced overall demand for center city residence, leading to white out-migration, and mostly to falling house prices
 - For every 1 Black moving in, around 2.7 White moved out
 - But: Cannot distinguish between racial bias and distaste for income level of new migrants

Causes of segregation: White flight

(Boustan, 2010)

Number of Black migrants leaving/entering South (by decade)



Source: Boustan (2017)

Causes of segregation: White flight

(Boustan, 2010)

- Paper examines the impact of Black migration from the US South on White residential decisions and housing demand in (northern) central cities
- *How can we interpret the results of the paper in light of the bid-rent theory?*
 - Which bid-rent curves do you expect to move, and under what assumptions?
 - Think about how changes in neighborhood demographics may affect White residents willingness to pay for proximity to center

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Dynamic models of segregation

(Schelling, 1971)

- Shows how individual preferences for neighborhood composition can lead to large-scale segregation patterns
 - Even when those preferences are relatively mild
- Early example of agent-based model, where individuals act depending on local conditions
- Mechanism: Individuals have a threshold preference for own-race neighbors
 - They want at least a certain % of neighbors of their same race
 - If lower, they move to a different neighborhood

Dynamic models of segregation

(Schelling, 1971)

The line of stars and zeros in Figure 1 corresponds to the odd and even digits in a column of random numbers. It turns out that there are 35 stars and 35 zeros, and they look reasonably 'random.' (There is no need to test for oddities and regularities; it is easier to replicate numbers.) We interpret these stars and zeros to be people spread out in a line, each concerned about whether his neighbors are stars or zeros.

We expect the stars and zeros to be evenly distributed in the large but unevenly in the small. If stars and zeros are content to live together in a ratio of about 50:50, each finds himself in a satisfactorily mixed neighborhood if he defines his neighborhood as a long enough stretch of the line. If instead everybody defines 'his neighborhood'

0+000++0+00++00++0+0+0+00++00++0+0+00++0++00000++000+00++0+0+0

Fig. 1

as his own house and the two neighbors next to him, a quarter of the stars and zeros are going to be 'surrounded' by neighbors of opposite color. Satisfaction depends on how far one's 'neighborhood' extends.

Suppose, now, that everybody wants at least half his neighbors to be like himself, and that everyone defines 'his neighborhood' to include the four nearest neighbors on either side of him. A star wants at least four of his eight nearest neighbors to be stars; a zero wants at least four of his eight nearest neighbors to be zeros. Including himself, this means that he wants a bare majority, five out of the nine. (For those near the end of the line the rule is that, of the four neighbors on the side toward the center plus the one, two or three outboard neighbors, half must be like oneself.)

I have put a dot over each individual whose neighborhood does not meet his demands.

Dynamic models of segregation

(Schelling, 1971)

- High levels of intolerance lead to high segregation
- But (maybe surprisingly): High levels of segregation also emerge from *low* levels of intolerance
 - E.g., people happy with 60% diverse neighbors
- Individuals may self-segregate even when they have preference for mixed neighborhoods
- *What do you think happens if people are tolerant of diversity up to 40%? And 60%? And if the minority share increases?*
 - Model simulation [here](#)

Tipping and the dynamics of segregation (Card et al., 2008)

- Schelling (1971) model shows that extreme segregation can arise from social interactions in White preferences
 - Once minority share in a neighborhood exceeds a “tipping point,” all White population leaves
- Card et al. (2008) examine whether *real* neighborhoods exhibit tipping-like dynamics
 - i.e., whether small increase in minority population in the neighborhood leads to large departures of majority population
- Use US census data on racial composition of neighborhoods

Tipping and the dynamics of segregation (Card et al., 2008)

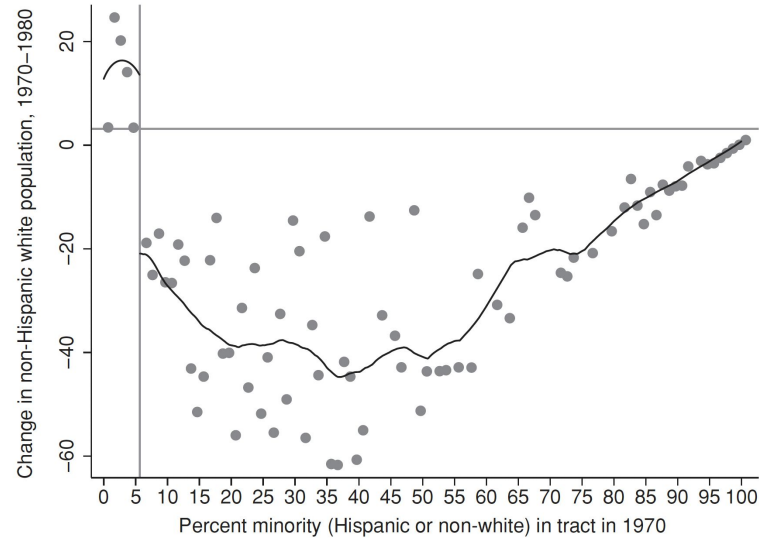


FIGURE I
Neighborhood Change in Chicago, 1970-1980

Tipping and the dynamics of segregation (Card et al., 2008)

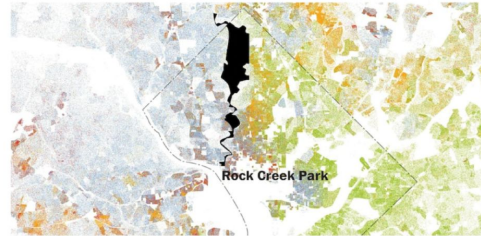
- Key findings: In many US neighborhoods, there is a tipping point minority share (10-20%) at which White residents begin moving out at faster rate
 - Heterogeneous differences across locations and minorities
 - Small differences in minority share around tipping point leads to large differences in neighborhood composition
- Results in line with Schelling's predictions
- *Implications for real estate?*

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Lines of segregation (Bagagli, 2025)

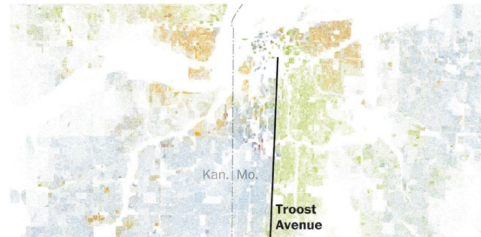
(a) Washington, D.C.



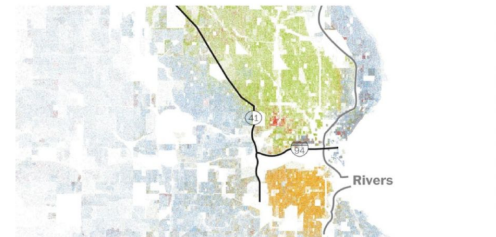
(b) Shreveport, La.



(c) Kansas City, Mo.



(d) Milwaukee, Wi.



1 dot = 1 person ● White ● Black ● Asian ● Hispanic ● Other

Source: The Washington Post, 2015 (2010 census data)

The (Express)Way to Segregation

(Bagagli, 2025)

Q: How do highways affect racial segregation within cities?



(a) 1938



(b) 2020

The (Express)Way to Segregation

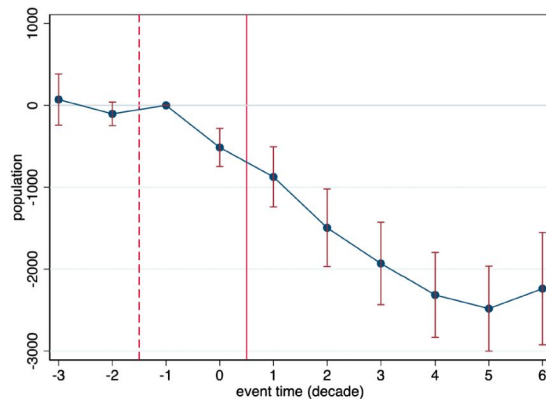
(Bagagli, 2025)

- Two main channels
 - 1 Disamenity channel
 - Highways create pollution, noise, other disamenities nearby
 - In city centers, house value goes down near highways
 - (Poor) Black households move in, (rich) White households out
 - 2 Barrier effect channel
 - Highways are physical barriers within cities
 - Technology that enables people to segregate, by *physically* separating themselves from the other group
- Other channels linking highways to segregation?

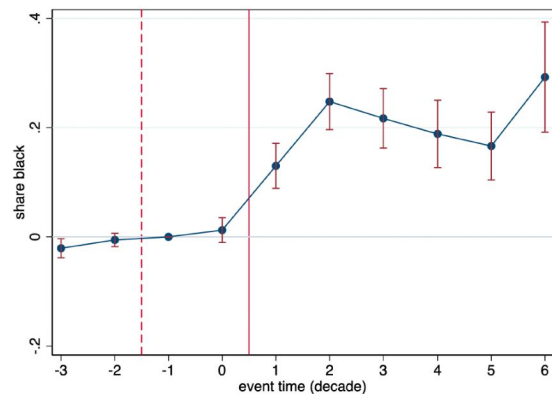
Disamenity channel (Bagagli, 2025)

$$y_{it} = \alpha_i + \gamma_t + \sum_{j=-3}^6 \beta_j D_i \times T_{i,t=t^*+j} + \epsilon_{it}$$

(a) Outcome (y_{it}): Residential population



(b) Outcome (y_{it}): Share Black



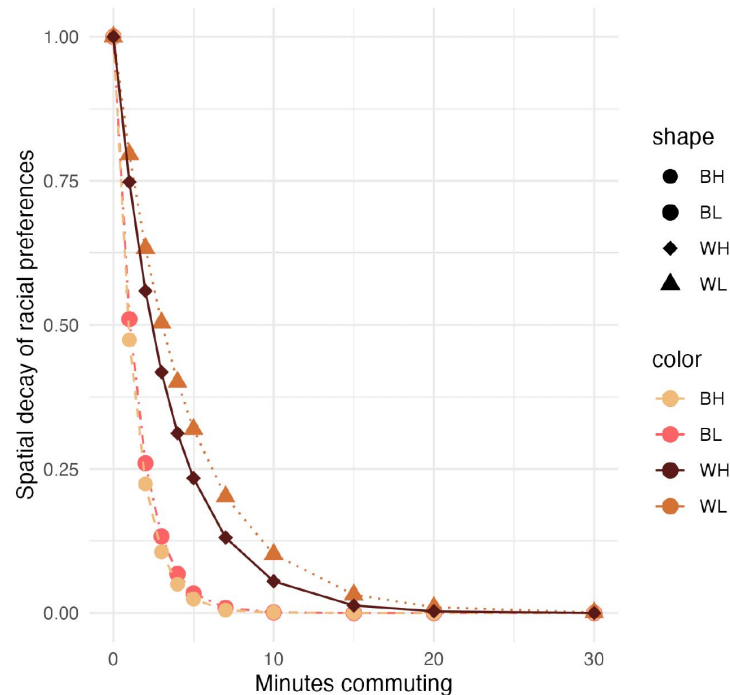
Other results (Bagagli, 2025)

- House value, share college educated (proxy for income) go down in proximity to highways
- Sizable barrier effects
 - 10 p.p. increase in exposure to Black leads to 3 p.p. increase in share Black living in the neighborhood
- Quantitative urban model
 - Estimate racial preferences (e.g., spatial decay and elasticity of density of same vs. other race in nearby areas)
 - Counterfactuals: If we put highways underground, segregation in Chicago would go down by 17%

Spatial decay of racial preferences

(Bagagli, 2025)

- For Black, residential location choice depends on race of neighbors living up to 10 min away
- For White, up to 20 min away



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Consequences of segregation

- Segregation increases poverty rate (Ananat, 2011) and reduces upward mobility (Chyn et al., forthcoming)
- Great Migration from South associated with segregation and lower mobility rates, especially for Black males (Derenoncourt, 2022)
- Worse labor market outcomes (Cutler and Glaeser, 1997)
- But also: Segregation linked with higher crime, mortality, exposure to pollutants,...

Role of ethnic enclaves

- What is the causal effect of living in an ethnic enclave on immigrant economic outcomes? (Edin et al., 2003)
- How can we estimate this?
- Naive: Regress outcomes of interest on whether immigrant lives in ethnic enclave or not
 - Comparing outcomes of those inside and outside of it
 - Issues? Selection!
 - Solution? Use natural experiment from Swedish refugee assignment policy: Refugees assigned randomly to municipalities, creating (exogenous) variation in size of ethnic enclaves
- Results: Enclaves improve short-term job prospects due to social network, but in long-run hinder assimilation

Summary

- The choice of house embeds choice of neighborhood too
 - Nhbd demographics influence housing demand and prices
- Segregation in the US peaked in 1970s, declining since then but strong persistence
 - Collective exclusion contributed to early segregation
 - White flight prominent throughout (to suburbs after 1940s)
 - Urban forms can contribute to segregation in cities
- Segregated cities perform poorly on important socio-economic outcomes
 - Poverty, crime, upward mobility, educational attainment,...
- Living in large enclaves can be beneficial at first, but may hinder assimilation and economic outcomes

Next week: Gentrification and Suburbanization

- Determinants of long-run trends
 - Relative transportation costs by income groups
 - Lifecycle of buildings
 - Amenities
 - Endogenous gentrification
- Welfare dimension
 - Winners and losers
 - Role of mobility costs
- Consequences of gentrification

Thank you!

Readings and references

Required readings

- Boustan, L. P. (2010) Was postwar suburbanization “white flight”? Evidence from the black migration. *The Quarterly Journal of Economics*, 125(1), 417-443.
- Cutler, D. M., Glaeser, E. L., and Vigdor, J. L. (1999). The rise and decline of the American ghetto. *Journal of political economy*, 107(3), 455-506.
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Other readings and references

- Aaronson, D., Hartley, D., and Mazumder, B. (2021). The Effects of the 1930s HOLC “Redlining” Maps. *American Economic Journal: Economic Policy*, 13 (4), 355-392.
- Ananat, E. O. (2011). The wrong side (s) of the tracks: The causal effects of racial segregation on urban poverty and inequality. *American Economic Journal: Applied Economics*, 3(2), 34-66.
- Bagagli, S. (2025). The (Express)Way to Segregation: Evidence from Chicago. Available at: https://drive.google.com/file/d/1f45RDOze8EiGvIVv1chgOp3cOPu5i_qn/view
- Boustan, L. P. (2017) *Competition in the promised land: Black migrants in northern cities and labor markets*. Princeton University Press.
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Readings and references

- Chyn, E., Haggag E., and Stuart B. A. (forthcoming). The Effects of Racial Segregation on Intergenerational Mobility: Evidence from Historical Railroad Placement. Forthcoming, American Economic Journal: Applied Economics
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- Schelling, T. C. (1971). Dynamic models of segregation. Journal of mathematical sociology, 1(2), 143-186.

Additional resources

